PAPER W: VERIFICATION RECORD

VICO BONFIOLI

This document supports Paper W (`erdos634_v3W2.tex`); the proof there does not need it. It records how each statement was checked, the older complex-coordinate proof, and where data and Lean files are. No reader outside this project has checked any of it.

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1. What is checked how

statement	checked by
Theorem W, W half	Lean (`Draft.ThmW.thmW`, 19 helper modules, aggregate `Erdos634.AllThmW`); and the hand proof of the paper (not Lean)
Frame (coordinates of the points, metric, five directions)	hand; exact arithmetic against the older complex description, all coprime e, f with $f \leq 24$, $M < 3f$ (sampled M for $f \leq 40$), 40 624 cases (e, f, M, t, r, q): no mismatch (`code/check_thmW_frame.py`, block A)
The 24 edge identities, parallelism, cut lines	hand; exact: 18 276 cases (block B); and 3000 random integer parameter sets (also outside (K)) re-checked 2026-10-19 for the rewrite: all 24 identities and the two parallelogram identities hold identically
The 22 turns and their factorisation into the nine factors	computer algebra (sympy), 2026-10-19, a scratch script, not shipped; not a saved tool; a Rust check (`anvil check/chain`) is planned
Chain identity (five line equations)	hand; collinearity of the points on the five lines identically in e, f, M, t, r, q (block A, exact)
Layout: containment, separation, convexity, orientation, areas	hand; exact on 201 886 triples (e, f, M), $f \leq 60$, $M < 5f$: no failure under (H1), (H2), none passes outside (block D)
(K) $\Leftrightarrow$ the sixteen sign conditions of the older proof; layout exists iff $dn \geq e{+}1$; canonical triple works whenever any works	hand; exact enumeration of all (t, r, q) for $f \leq 12$, $M < 3f$, 17.9 million triples (block E)
Lemma H (both directions)	hand; exact: coprime $2 \leq e \leq f - 2$, $f \leq 300$, $M \leq 3f$, 16.2 million triples (block G); Lean version `Chain.W.lemmaH` (unaudited, no `sorry`)
Certificates of Proposition 4.5 of the paper	hand; exact on all 982 658 admissible (e, f, M) with $f \leq 120$, $M \leq 3f$; the 27 322 non-admissible ones each violate one of (K)
Count identity (six counts add to $M^2\kappa$)	computer algebra (block H, sympy)
Chain lemma (Lean)	`Draft/ChainKap`, `ChainTri`, `ChainCover`: triangles; `ChainW`: the six pieces split into ten triangles; `ChainParams`: Lemma H and canonical parameters. Unaudited, no `sorry`; not linked to the packing statement (stage 3 open), so no `WTilable` derived from them
Packing: convex pieces, each tiled, disjoint interiors, area sum = area of target $\Rightarrow$ tiling	Lean (general theorem in the library)
Explicit tilings (414, 656, 639, 801, 3950, 13 888, 57 232 tiles)	exact checker on exact coordinates: congruence, disjointness, area, hull a triangle (not Lean)
26 138 new tile counts	program count against the census of the companion (not Lean); each value is an instance of Theorem W

(For the canonical triple, $C_3 = \rho + f\delta$, $C_4 = f - \rho$, $C_6 = dn - e$, $C_5 = dn - e + f(M - n)$, $C_7 = dn - f\delta$.) Consistency with the exclusions: (H1), (H2) imply $M(f - e) > e$, $Me \geq f$ and $2Me \geq f$, so no instance of the theorem contradicts (I1) or (I2).

2. The older proof: complex coordinates and cross-product tables

The first version of Paper W and the Lean proof use complex numbers with $O = 0$ and OB the positive real axis. With $D = 4f^2 - e^2$ write $\xi + i\sqrt{D}\,\zeta$ as $[\xi, \zeta]$ and put

$$\varepsilon = e^{i\alpha} = \left[\tfrac{\kappa}{2f^2}, \tfrac{e}{2f^2}\right], \quad \eta = e^{i\beta} = \left[\tfrac{e(3f^2-e^2)}{2f^3}, \tfrac{b}{2f^3}\right], \quad \omega = \overline{\varepsilon\eta} = e^{-i(\alpha+\beta)},$$

with $b = f^2 - e^2$, $\varepsilon^3\eta^2 = -1$, $a + b\omega = c\bar{\eta}$, $f\omega + e\varepsilon = f\bar{\eta}$. Then $P = fbM\varepsilon$, and the frame of the paper is $z = P - \xi\,\varepsilon + y\,\omega$. The points are $B = fbM$, $A = f^3M\varepsilon^2$, $P_2 = b(fM - et)\varepsilon$, $P_3 = P - eat\,\omega$, $Q = P + fbt\,\omega$, $R = P + fbr\,\omega$, $C' = bv\,\varepsilon$, $B' = cv$, $V_2 = P + f^3t\,\bar{\eta} - ef^2r$, $X_4 = P - fbq\,\varepsilon$, $U_1 = P - ef^2t + ef^2q\,\bar{\eta}$, $U_2 = P - ef^2r + ef^2q\,\bar{\eta}$ (exact check against the frame: block A). The Lean proof shows that each cross product needed for containment, separation and convexity (about 100) is a positive constant times a product of sixteen named factors

$$C_2 = r - t,\ C_3 = eM - ft,\ C_3' = fM - et,\ C_4 = fr - eM,\ C_5 = fM - er,\ C_6 = fq - er,\ C_6' = fq - et,$$
$$C_7 = ft - eq,\ C_8 = f^2t - e^2r,\ C_9 = fr - eq,\ C_{10} = M - q,\ E_1 = bM + eft,\ E_2 = eMf + bt,\ E_3 = br + e^2t,$$
$$L_A = bfM - e\kappa t + f^3q,\ L_B = bfM - e\kappa r + f^3q,\ L_C = efbM - e^2\kappa r + f^4t,$$

all positive for the canonical triple; the tables are generated by `python3 code/check_thmW_algebra.py 60 -tex tables.tex`. Values for three examples (all positive):

(e,f,M)	C_2	C_3	C_3'	C_4	C_5	C_6	C_6'	C_7	C_8	C_9	C_{10}	L_A	L_B	L_C
$(2,5,3)$	1	1	13	4	11	1	3	3	17	8	2	348	256	887
$(3,5,4)$	1	2	14	3	11	1	4	4	23	9	2	324	201	1103
$(2,5,5)$	2	5	23	5	19	4	8	1	13	11	3	683	499	1123

3. Data, Lean index, reproduction

Thirteen witnesses with $M < f$ found by the first computer search (now instances of Theorem W; each a Lean theorem in `Draft/DebtBWTilable`); tiles of W_M and T_M:

(e,f,M)	W_M	T_M	(e,f,M)	W_M	T_M
$(3,5,4)$	656	1056	$(7,12,4)$	3824	6128
$(4,7,4)$	1312	2096	$(7,10,5)$	3775	6275
$(5,8,4)$	1648	2672	$(7,9,6)$	4068	6984
$(5,7,5)$	1825	3050	$(8,11,5)$	4450	7475
$(5,11,3)$	1953	3042	$(9,11,7)$	7889	13 818
$(5,9,4)$	2192	3488	$(6,11,4)$	3296	5232
$(7,11,4)$	3088	5024			

This search was bounded by its own parameter range (it did not reach $W_3(2,5)$). *Shipped tilings:* directory `data/witnesses/v3_exact/`, files `tiling_<count>_<kind>_<e>_<f>_M<M>.exact.txt.gz`, kinds `W` (37 files) and `T` (15); each starts with a comment line, then one tile per line in exact coordinates $(X, Y\sqrt{D})$. *Lean index* (namespace `Erdos634.`, aggregate `AllThmW`): `Draft.ThmW.thmW`; modules `ThmWLib`, `Lib2`, `Hyp`, `Sc`, `Pts`, `Ineq0`–`4`, `Geo0`–`3`, `Core`, `Nat`, `Params`, `Final`; instances `inst_414`, `inst_656`, `inst_639`, `inst_801`; counts `Draft.DebtB.counts_of_wTilable`; glue `CollarG.betaTilable_of_wTilable`; ladder `Subdivision.ladderDissection`.

```
python3 code/check_thmW_frame.py ALL 60                 # frame, layout, equivalences
python3 code/check_thmW_algebra.py 60 --tex tables.tex  # older algebra + tables
python3 code/lakeserial.py lean build Erdos634.AllThmW  # one module at a time
BTE=rust/bonfioli_tiling_engine/target/release/bte
$BTE verify data/witnesses/v3_exact/tiling_414_W_2_5_M3.exact.txt.gz
```